AEN 340: Phonetic and Phonological Analysis

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PAPER 1

KENYATTA UNIVERSITY AND PHONOLOGICAL ANALYSIS

1st Semester 2024/2025 | Time: 2 hours

INSTRUCTIONS: Answer Question ONE and any other TWO questions.

QUESTION ONE (COMPULSORY)

a) Using the parameters of airstream mechanism, voicing, place of articulation, cavity and manner of articulation, describe the following sounds. (10 marks)

i. /ɸ/

ii. /b/

iii. /ɓ/

iv. /p'/

v. /ŋ/

Answer

Sound	Airstream	Voicing	Place	Cavity	Manner
i. /ɸ/	Pulmonic egressive	Voiceless	Bilabial	Oral	Fricative
ii. /b/	Pulmonic egressive	Voiced	Bilabial	Oral	Plosive (stop)
iii. /ɓ/	Glottalic ingressive	Voiced	Bilabial	Oral	Implosive
iv. /p'/	Glottalic egressive	Voiceless	Bilabial	Oral	Ejective plosive

Sound	Airstream	Voicing	Place	Cavity	Manner
v. /ŋ/	Pulmonic egressive	Voiced	Velar	Nasal (velum lowered)	Nasal stop

b) Give any five allophones of the phoneme /t/ and for each allophone, explain the environment in which it is produced. (10 marks)

Answer

An allophone is a variant of a phoneme that does not change meaning. The phoneme /t/ has these:

1. **[tʰ] aspirated /t/:** occurs word-initially and at the start of a stressed syllable, before a vowel. Examples: *top* [tʰɒp], *attack* [əˈtʰæk].
2. **[t] unaspirated /t/:** occurs after /s/ in the same syllable. Examples: *stop*, *star*.
3. **[t̚] unreleased /t/:** occurs word-finally or before another stop, when the closure is held without audible release. Examples: *cat*, *that boy*, *hot dog*.
4. **[ʔ] glottal stop:** replaces /t/ before a syllabic nasal or word-finally in some accents. Examples: *button* [bʌʔn̩], *bat* [bæʔ].
5. **[ɾ] alveolar flap:** occurs between vowels, with the stress before it, in American English. Examples: *butter*, *water*.

Note; Other allophones are the dental [t̪] before /θ/ (*eighth*), and the labialised [tʷ] before /w/ (*twin*).

QUESTION TWO

a) Discuss the following distinctive oppositions. (12 marks)

i. Privative

ii. Bilateral

iii. Equipollent

iv. Gradual

Answer

Distinctive oppositions were proposed by Trubetzkoy (Prague School) to classify how phonemes contrast with each other.

i. **Privative opposition:** one member of the pair has a feature (marked) and the other lacks it (unmarked). The contrast is presence versus absence. Example: /p/ and /b/ differ in voicing: /b/ is voiced and /p/ is not. Likewise /t/ and /n/ (nasality present or absent).

ii. **Bilateral opposition:** the features common to the two members are shared by no other phoneme in the system, so the pair is a unique pair. Example: /p/ and /b/ in English are the only bilabial plosives, so the basis of comparison (bilabial and plosive) applies to that pair alone. The opposite is a **multilateral** opposition, such as /p/ and /t/, where the common features (voiceless plosive) are shared by /k/ also.

iii. **Equipollent opposition:** the two members are logically equal, each having its own positive feature, and neither is merely the absence of the other. Examples: /p/ (bilabial) and /t/ (alveolar); /f/ (labiodental) and /θ/ (dental).

iv. **Gradual opposition:** the members differ in the degree or gradation of the same property. Example: the vowels /iː/, /e/ and /æ/ differ in tongue height (close, mid, open).

b) Describe the following vowel sounds according to length, horizontal tongue height, vertical tongue height and lip position. (8 marks)

i. /æ/ ii. /ʌ/ iii. /ɑː/ iv. /iː/

Answer

Vowel	Length	Horizontal tongue position	Vertical tongue height	Lip position
i. /æ/	Short	Front	Near-open (open-mid to open)	Unrounded (spread)
ii. /ʌ/	Short	Central to back (centralised)	Open-mid	Unrounded (neutral)
iii. /ɑː/	Long	Back	Open	Unrounded (neutral)
iv. /iː/	Long	Front	Close	Unrounded (spread)

QUESTION THREE

a) With examples, explain the following phonological processes. (12 marks)

i. Labialisation ii. Voicing assimilation iii. Insertion iv. Metathesis

Answer

i. **Labialisation:** a sound acquires lip-rounding (a secondary articulation) from a neighbouring rounded segment. Examples: /s/ becomes [sʷ] in *swim*; /ʃ/ becomes [ʃʷ] in *shoe*; /t/ becomes [tʷ] in *twin*.

ii. **Voicing assimilation:** a sound takes on the voicing value of a neighbouring sound. Examples: the plural morpheme /z/ is voiceless [s] after voiceless sounds (*cats* [kæts], *books*) but stays voiced [z] after voiced sounds (*dogs* [dɒgz]). Also *have to* becomes [hæf tə], where /v/ devoices before /t/.

iii. **Insertion (epenthesis):** a segment is added to a word, often to break up a difficult cluster. Examples: *film* becomes [fɪləm] in some accents; Swahili *shule* from English *school* and *kalabu* from *club*; *athlete* becomes [æθəliːt].

iv. **Metathesis:** two sounds in a word swap positions. Examples: *ask* becomes *aks*; *relevant* becomes *revelant*; Old English *brid* became *bird*; *wasp* becomes *waps*.

b)

(i) Using the greater than (>) symbol to stand for 'more sonorous than', give the sonority hierarchy of the following sounds. (4 marks)

/l, r, m, uː, t, b, k, æ, z, iː, ɜː, e, ʃ, g/

Answer

/æ/ > /e/, /ɜː/ > /iː/, /uː/ > /l/, /r/ > /m/ > /z/ > /ʃ/ > /b/, /g/ > /t/, /k/

That is: low vowel > mid vowels > high vowels > liquids > nasal > voiced fricative > voiceless fricative > voiced stops > voiceless stops.

(Sounds on the same rank, such as /b, g/ and /t, k/, are equal. In some charts /r/ ranks above /l/.)

(ii) Explain the relationship between sonority and the syllable. Use examples. (4 marks)

Answer

Sonority is the relative loudness (**inherent carrying power**) of a sound compared with others of the same length, stress and pitch. The syllable is built around it:

- The **nucleus (peak)** of a syllable is the most sonorous element, usually a vowel (or a syllabic consonant).
- By the **Sonority Sequencing Principle**, sonority **rises** from the margin (onset) towards the peak and **falls** from the peak towards the end (coda).
- Example: *plant* /plɑːnt/. Sonority goes /p/ < /l/ < /ɑː/ > /n/ > /t/, so /ɑː/ is the peak, /pl/ is the onset and /nt/ is the coda.
- Example: *strike* /straɪk/. The /s/ is an exception, since /s/ + stop before a rise is allowed in English (s-extrasyllabicity), but the peak is still /aɪ/.
- Syllable boundaries fall at sonority troughs: *Atlas* /ˈæt.ləs/ divides at the dip between /t/ and /l/.

QUESTION FOUR

With examples, differentiate the following: [20 marks]

(i) Combinatory and Facultative variants
(ii) Heavy and light syllables
(iii) (iii) Plosives and nasal
(iv) (iv) Stress-timed language and tonal language
(v) (v) Regressive and progressive assimilation

Answer

(i) Combinatory and facultative variants

- **Combinatory (positional) variants** are allophones determined by the phonetic environment. They are in complementary distribution: where one occurs the other cannot. Example: [p^h] occurs word-initially (*pin*) and [p] after /s/ (*spin*).
- **Facultative (free) variants** can occur in the same environment without any change in meaning, depending on the speaker's choice or style. Example: word-final /t/ in *cat* may be released [kæt^h] or unreleased [kæt̚].

(ii) Heavy and light syllables

- A **heavy syllable** has a branching rhyme: a long vowel or diphthong, or a short vowel plus a coda consonant. Examples: *beat* /biːt/, *bat* /bæt/, *time* /taɪm/.
- A **light syllable** has a short vowel with no coda (CV), so the rhyme does not branch. Examples: the first syllable of *butter* /bʌ/, and /tə/ in *today*. Heavy syllables attract stress.

(iii) Plosives and nasals

- **Plosives:** the velum is raised, so air cannot pass through the nose. There is complete oral closure, pressure builds up, and the air is released with an explosion (burst). Examples: /p, b, t, d, k, g/. They are obstruents.

- **Nasals:** the oral closure is the same, but the velum is lowered so air escapes through the nose. Examples: /m, n, ŋ/. They are sonorants and are usually voiced.
- /b/ and /m/ share the same place (bilabial) and differ in whether air escapes through the nose.

(iv) Stress-timed language and tonal language

- A **stress-timed language** (English, German) has stressed syllables occurring at roughly equal time intervals, and the unstressed syllables between them are shortened or reduced. Stress is signalled by loudness, length and pitch. Example: *The CAT sat on the MAT* takes about as long as *The CAT has been SITting on the MAT*, because unstressed syllables are squeezed.
- A **tonal language** (Igbo, Yoruba, Mandarin, Luganda, Dholuo) uses the pitch level or contour on each syllable to distinguish word meaning or grammar. Example (Igbo): *àkwà* "bed", *ákwá* "cry", *àkwá* "egg", *ákwà* "cloth". Mandarin: *mā* "mother", *mǎ* "horse".

(v) Regressive and progressive assimilation

- **Regressive (anticipatory) assimilation:** a sound is influenced by the sound that follows it. Example: *in + possible* becomes *impossible* (/n/ becomes /m/ before bilabial /p/); *ten boys* becomes [tem bɔɪz].
- **Progressive (perseverative) assimilation:** a sound is influenced by the sound that precedes it. Example: plural /z/ becomes /s/ after a voiceless sound (*cats*); *dogs* keeps /z/ after a voiced sound.

QUESTION FIVE

(a) Explain any FOUR functions of each of the following speech organs in the production of sounds. Use examples. [16 marks]

(i) Glottis

(ii) Velum

Answer

(i) The glottis (the space between the vocal folds)

1. **Voicing:** the folds vibrate when closed together and air passes through them. Voiced /b, d, g, z/ contrast with voiceless /p, t, k, s/.
2. **Glottal sounds:** closure produces the glottal stop [ʔ] (*button* [bʌʔn̩]) and narrowing produces the glottal fricative /h/ (*hat*).
3. **Initiating the glottalic airstream:** the closed glottis is raised or lowered to produce ejectives [p’, t’, k’] and implosives [ɓ, ɗ].
4. **Phonation types and pitch:** whisper, breathy voice (murmur) and creaky voice, and the rate of vibration controls the pitch used for tone and intonation (Igbo *àkwà* vs *ákwá*).

(ii) The velum (soft palate)

1. **Raised (velic closure):** it closes off the nasal cavity so air passes through the mouth to produce oral sounds such as /p, b, s, a/.
2. **Lowered:** it opens the nasal cavity to produce nasal consonants /m, n, ŋ/.
3. **Place of articulation:** it is the passive articulator for velar sounds /k, g, ŋ/ (the back of the tongue touches it).
4. **Nasalisation and clicks:** vowels next to nasals are nasalised (*man* [mæ̃n]), and the velum with the back of the tongue provides the closure for the velaric airstream in clicks (Zulu, Xhosa).

(b) With the aid of the features given below, use a feature matrix to describe the sounds below.

[voice] [syll] [cons] [son] [cont] [ant]

(i) [k]
(ii) (ii) [θ]

Answer

Feature	(i) [k]	(ii) [θ]
[voice]	–	–
[syll]	–	–
[cons]	+	+
[son]	–	–
[cont]	–	+
[ant]	–	+

PAPER 2

KENYATTA UNIVERSITY

EXAMINATION FOR THE DEGREE OF BACHELOR OF EDUCATION

AEN 340: PHONETIC AND PHONOLOGICAL ANALYSIS

1st Semester 2022/2023 | Time: 2 hours

INSTRUCTIONS: Answer Question ONE and any two others.

QUESTION ONE (COMPULSORY)

a. Provide the information required: (5 marks)

i. The phonetic symbol for a palatalized voiced velar stop.

ii. The phonetic symbol for a lateral that is devoiced.

iii. The phonetic symbol for a bilabial click.

iv. The phonetic symbol for any sound produced with the egressive glottalic airstream mechanism.

v. The phonetic symbol for the diphthong in the words **'face'** and **'bay'**.

Answer

i. **[gʲ]**: the voiced velar stop with the **palatalisation** diacritic.
ii. **[l̥]**: the lateral with the voiceless ring diacritic (a voiceless lateral fricative is [ɬ]).
iii. **[ʘ]**: the bilabial **click**.
iv. **An ejective**, marked with an apostrophe, for example [p'], [t'] or [k'].
v. **/eɪ/**

b. For each sound, pick out the wrongly marked features from the following table. (10 marks)

The table has the sounds [g], [θ], [ð], [j] and [h] across the top, and the features [SONORANT], [NASAL], [ANTERIOR], [CORONAL], [HIGH], [BACK] and [CONTINUANT] down the side.

Feature	[g]	[θ]	[ð]	[j]	[h]
[sonorant]	–	–	–	+	–
[nasal]	–	–	–	–	–
[anterior]	–	+	+	–	–
[coronal]	–	+	+	–	–
[high]	+	–	–	+	–
[back]	+	–	–	–	–
[continuant]	–	+	+	+	+

***Note**: Some charts mark [j] as [+coronal]. You can use the convention in the notes we've been given in class.*

c. Study the following words then answer the two questions that follow:

quick, sweet, tweet, shoe

i. Explain the assimilatory process that is common to all the above words. (2 marks)

Answer

The common process is **labialisation (rounding assimilation)**. The consonant before a rounded segment (/w/ in *quick, sweet, tweet*, and the rounded vowel /uː/ in *shoe*) takes on lip-rounding. The rounding of the following sound is anticipated, so the assimilation is regressive.

ii. Use any word from the words above to write a phonological rule to represent the identified process in (i) above. (3 marks)

Answer

Using *sweet* /swiːt/: the consonant /s/ is unrounded, but it becomes rounded before /w/.

/s/ → [s^w] / ___ /w/

General form: **C → [+round] / ___ [+round]**

In feature notation: [+cons, −round] → [+round] / ___ [+round]. So *sweet* is pronounced [sʷwiːt].

QUESTION TWO

Use the following sentence to answer the questions that follow.

Father caught green birds.

a. Provide a phonemic transcription for the sentence. (5 marks)

Answer

/ˈfɑːðə kɔːt griːn bɜːdz/

b. Plot all the vowels in the sentence on a vowel trapezium. (5 marks)

Answer

The vowels in the sentence are **/ɑː/, /ə/, /ɔː/, /iː/ and /ɜː/.** Position them on the trapezium (front on the left, back on the right, close at the top, open at the bottom) as follow **(and by the way I believe we all know how to draw and plot on the vowel trapezium coz this was taught in AEN 210 & AEN 212 by Dr. Itumo)**

Vowel	Word	Horizontal	Vertical	Lips
/iː/	green	Front	Close	Spread
/ɜː/	birds	Central	Open-mid	Neutral
/ə/	father (2nd syllable)	Central	Mid (schwa)	Neutral
/ɔː/	caught	Back	Open-mid	Rounded
/ɑː/	father (1st syllable)	Back	Open	Unrounded

So, on your drawing: put /**i**ː/ at the top left corner, /**ɔ**ː/ on the right side below the middle line, /**ɑ**ː/ at the bottom right, and /**ɜ**ː/ and /**ə**/ in the centre (/**ɜ**ː/ a little lower than /**ə**/).

c. Provide a feature matrix for the first five consonants in the sentence using the following features: (10 marks)

i. Syllabic

ii. Back

iii. Nasal

iv. Voice

Answer

The first five consonants are **/f/, /ð/, /k/, /t/ and /g/.**

Feature	/f/	/ð/	/k/	/t/	/g/
[syllabic]	–	–	–	–	–
[back]	–	–	+	–	+
[nasal]	–	–	–	–	–
[voice]	–	+	–	–	+

QUESTION THREE

Giving relevant examples for each type, discuss any four non-assimilatory processes. (20 marks)

Answer

Non-assimilatory processes change a sound without making it more like a neighbouring sound. Four of them are:

1. **Elision (deletion):** a sound is dropped. Examples: *next day* /nekst deɪ/ becomes /neks deɪ/; *friendship* becomes /frenʃɪp/; *comfortable* becomes /ˈkʌmftəbl/. In rule form: C → Ø / C ___ C.
2. **Epenthesis (insertion):** a sound is added, often to break up a cluster. Examples: *film* becomes [fɪləm]; *athlete* becomes [æθəliːt]; Swahili *shule* from *school*. In rule form: Ø → ə / C ___ C.
3. **Metathesis:** two sounds swap places. Examples: *ask* becomes *aks*; *relevant* becomes *revelant*; Old English *brid* became *bird*.

4. **Dissimilation:** two similar sounds in a word become less alike. Examples: Latin *peregrinus* became English *pilgrim* (r…r became l…r); *February* is sometimes said *Febuary*.

- Other processes, any of which may replace one above: **lenition (weakening)**, such as /t/ becoming the flap [ɾ] in *water* in American English; **coalescence (fusion)**, such as /t/ + /j/ becoming /tʃ/ in *don't you* [dəʊntʃuː]; and **strengthening**, **vowel reduction** and **linking /r/**.

QUESTION FOUR

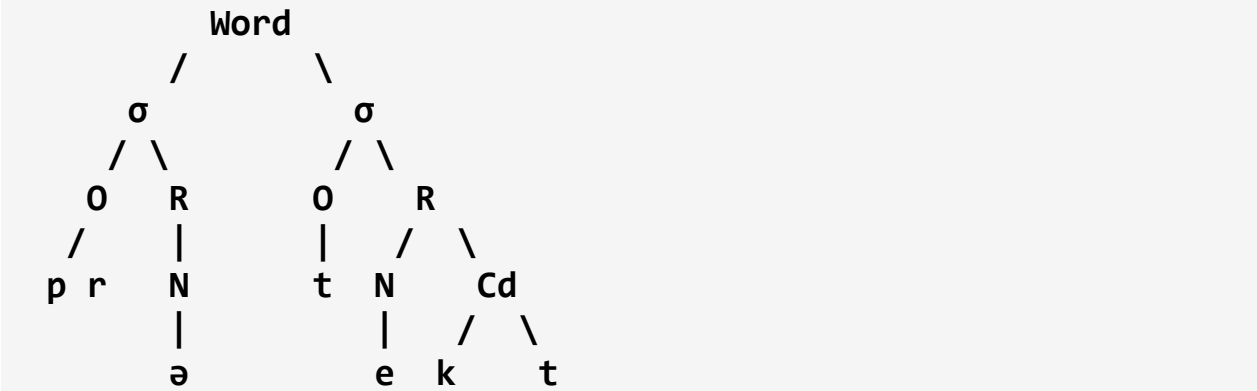
a. Draw syllable tree structures for the following words (10 marks)

'Protect', 'flour', 'prune', 'aye', 'prey'

Answer

Key: σ = syllable, O = onset, R = rhyme, N = nucleus, Cd = coda.

1. *protect* /prəˈtekt/ (two syllables)



- σ1: O = /pr/, R = [N = /ə/]
- σ2: O = /t/, R = [N = /e/, Cd = /kt/]

2. *flour* /flaʊə/ (two syllables; in rhotic accents /flaʊr/ is one syllable)

- σ1: O = /fl/, R = [N = /aʊ/]
- σ2: R = [N = /ə/] (no onset, no coda)

3. *prune* /pruːn/ (one syllable)

- σ: O = /pr/, R = [N = /uː/, Cd = /n/]

4. *aye* /aɪ/ (one syllable)

- σ: no onset, R = [N = /aɪ/], no coda

5. *prey* /preɪ/ (one syllable)

- σ: O = /pr/, R = [N = /eɪ/], no coda

Note: In each tree the syllable **(σ) branches into the onset (O) and the rhyme (R), and the rhyme branches into the nucleus (N) and the coda (Cd).**

b. Briefly demonstrate how tone can be used for prosodic distinction in any language of your choice. (10 marks)

Answer

Tone is the use of pitch on a syllable to distinguish word meaning (lexical tone) or grammatical function (grammatical tone). I use **Igbo**, which has two level tones (high ´ and low `), and a **downstep**.

Lexical tone. The same segments /akwa/ give four different words:

Word	Tone pattern	Meaning
àkwà	low-low	bed
ákwá	high-high	cry
àkwá	low-high	egg
ákwà	high-low	cloth

So pitch alone makes the difference, in the same way that a vowel or consonant would in English (*bat* vs *bit*).

Grammatical tone. In many Bantu languages, tone marks tense or case. Example from Luganda and other Bantu languages: a change in tone on a verb can separate the present tense from the past, with the consonants and vowels unchanged.

Other languages. Mandarin has four tones on /ma/: *mā* (high level) "mother", *má* (rising) "hemp", *mǎ* (falling-rising) "horse", *mà* (falling) "scold". Yoruba has three level tones: *ọkọ́* "spear", *ọkọ̀* "vehicle" and *ọkọ* "husband".

Tone differs from stress and intonation: in a tone language every syllable carries its own pitch, and changing it changes the word, whereas in English intonation changes the speaker's attitude or sentence type rather than the word's identity.

End of work by Ravive Omondi Ochieng'; 0742600411|0796156714|0728563575

GRACIAS AND ALL THE BEST COLLEAGUES!!!